DRAFT | SYNTHETIC DATA

Penetration Testing Report

Synthetic Training Organization

Engagement	TRAINING-001
Team	Training assessment team
Report date	2026-10-14
Document label	SYNTHETIC TRAINING DATA
Handling marking	ORCON//PROPIN//PII Red//FI

Training demonstration only. No real assessment, event permission, or submission is established.

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1 Introduction

Assessment objective

Demonstrate a structured assessment report using invented observations. No actual client, host, or network was assessed.

Scope

- Synthetic web application at training.example.test

- Demonstration observations only; no network actions were performed

Limitations

- No real systems, credentials, client data, or actual CPTC environment were tested.

- Candidate observations require human validation; this report is not an event submission.

- CVE/CVSS values are intentionally omitted because no public vulnerability match was established.

- Handling tags in this invented fixture demonstrate inheritance only; no actual personal or financial information is included.

Data handling

Report handling marking: ORCON//PROPIN//PII Red//FI

ORCON applies to all data and derived analysis. PROPIN marks internal systems, methods and procedures; PII marks employee/customer information; PII Red marks identity-theft-sensitive PII; FI marks financial information. Combined analysis retains all applicable restrictions; PII Red supersedes PII. Markings do not authorize disclosure or determine sensitivity from prose.

2 High-Level Summary and Recommendations

Two synthetic observations demonstrate the workflow: one confirmed training finding about unnecessary information exposure and one candidate requiring validation. The generator carries supplied evidence and remediation into a draft report; it makes no independent vulnerability judgments.

Finding summary

Total: 2 | Confirmed: 1 | Candidate: 1

Critical: 0 | High: 0 | Medium: 1 | Low: 1 | Info: 0

Severity method

Supplied business severity; matrix not configured. No likelihood/impact scoring is performed. CVSS, when supplied, is separate severity metadata.

Recommendations

- In the synthetic scenario, keep detailed diagnostics in protected logs and return a generic public error with a correlation ID.

- Validate the complete session flow before deciding whether the candidate cookie configuration needs remediation.

3 Methodology

Assessment approach

This report demonstrates importing structured findings from future team-script adapters. The observations and evidence below are invented training data. Analysts would verify actual findings against authorized evidence before submission.

Information gathering

The fixture author defined a synthetic web application and two invented response examples. No customer inventory or scope was inferred.

Service enumeration

Only synthetic response metadata was reviewed. No port scan, active request, or service discovery was performed.

Validation

The training analyst reviewed each fixture and retained the supplied confirmed or candidate status. No exploitation or access gain is claimed.

Cleanup and retest

No target state was changed in this exercise. Remediation and retest actions below are proposed checks for an authorized environment, not completed work.

4 Detailed Findings

F-001: Unnecessary internal details in a public error response

Handling marking

ORCON//PROPIN//FI

Business severity

MEDIUM - In the training scenario, internal paths and component details are exposed without authentication; no sensitive records or access gain were demonstrated.

Validation status

CONFIRMED

Affected assets

- training.example.test / application error handler

Observation

A synthetic error-response fixture includes an internal path and application component names. The training analyst marked this finding confirmed after reviewing the fixture. This status describes the exercise only.

Validation steps and results

1. 2026-10-14T14:00:00-04:00: reviewed the synthetic application response fixture.

2. Compared the response against the training requirement that public errors contain only a correlation ID.

Business impact

In an actual deployment, unnecessary implementation details could aid reconnaissance. The demonstration does not establish access to client records or an exploitation chain.

Remediation

Configure the application error handler to return a generic message and correlation ID. Keep diagnostic paths and stack traces in access-controlled application logs. Add regression coverage for unauthenticated error responses.

Retest plan

1. In an authorized test environment, trigger the same error and verify the response contains only the generic message and correlation ID.

2. Confirm authorized operators can still correlate the ID with protected diagnostic logs.

Evidence E-001

Handling marking: ORCON//PROPIN//FI

Observed: 2026-10-14T14:00:00-04:00 | Collector: Training analyst

Source: synthetic-response-fixture 1.0

Normalized evidence summary: Invented response excerpt: error at /opt/training/app/handler.py. No actual host or file was accessed.

Private artifact reference (not embedded or verified): evidence/synthetic-error-response.txt

F-002: Candidate: incomplete session-cookie configuration

Handling marking

ORCON//PII Red

Business severity

LOW - A synthetic header suggests a missing attribute, but the full session flow and compensating controls have not been validated.

Validation status

CANDIDATE

Affected assets

- training.example.test / session creation

Observation

A training fixture suggests that one session-cookie attribute may be missing. The excerpt alone does not prove a vulnerability or an exploitable session flow.

Validation steps and results

1. 2026-10-14T14:10:00-04:00: reviewed only the synthetic header excerpt.

2. Full session behavior, transport settings, and compensating controls remain untested.

Business impact

If confirmed in context, weaker cookie handling could reduce protection against certain session attacks. No session compromise is claimed.

Remediation

First validate the complete session flow and identify the intended browser security controls. If a required cookie attribute is missing, configure it in the session middleware and verify application compatibility.

Retest plan

1. Review session creation and renewal responses in the authorized environment.

2. Verify the intended attributes persist throughout the flow and document any compensating controls.

Evidence E-002

Handling marking: ORCON//PII Red

Observed: 2026-10-14T14:10:00-04:00 | Collector: Training analyst

Source: synthetic-header-fixture 1.0

Normalized evidence summary: Invented header excerpt: Set-Cookie: training_session=[REDACTED]; Path=/. This is a fabricated value, not a credential.

5 Appendices

Evidence register

E-001 | Finding F-001 | 2026-10-14T14:00:00-04:00 | Collector: Training analyst | Source: synthetic-response-fixture 1.0

E-002 | Finding F-002 | 2026-10-14T14:10:00-04:00 | Collector: Training analyst | Source: synthetic-header-fixture 1.0

Review record

Human review pending.

Review labels are local metadata. Inspect the actual artifacts before any authorized use.

Evidence files are referenced, never opened. Supplied hashes do not verify the observations; the manifest identifies the input and output files.

Generator 0.4.0 | Input schema 1.2